

Quantum Field Theory – Homework 2

Name: _____ Due Date: 2026.05.12

The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.

Problem 1. Canonical Structure and the Failure of Commutator Quantization (35 points)

Consider the free Dirac action

$$S = -i \int d^4x \bar{\psi}(\gamma^\mu \partial_\mu - m)\psi.$$

- Rewrite the Lagrangian density in the form

$$\mathcal{L} = i\psi^\dagger \partial_0 \psi - i\bar{\psi}(\gamma^i \partial_i - m)\psi.$$

- Compute the canonical momenta conjugate to ψ and $\psi^\dagger$, and show that

$$\pi_\psi = i\psi^\dagger, \quad \pi_{\psi^\dagger} = 0.$$

Briefly explain why this means the Dirac theory is a constrained system.

- Using the identification of $\psi^\dagger$ as the momentum conjugate to ψ , derive the Hamiltonian density

$$\mathcal{H} = i\bar{\psi}(\gamma^i \partial_i - m)\psi.$$

- Suppose one tries to quantize the field using *commutators* and the mode expansion

$$\psi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \sum_{s=1,2} [a^{(s)}(\mathbf{k}) u^s(k) e^{ik \cdot x} + b^{(s)}(\mathbf{k}) v^s(k) e^{-ik \cdot x}].$$

Show that this leads schematically to oscillator relations of the form

$$[a^{(s)}(\mathbf{k}), a^{(r)\dagger}(\mathbf{k}')] \sim \delta^{sr} \delta^{(3)}(\mathbf{k} - \mathbf{k}'), \quad [b^{(s)}(\mathbf{k}), b^{(r)\dagger}(\mathbf{k}')] \sim \delta^{sr} \delta^{(3)}(\mathbf{k} - \mathbf{k}').$$

- Show that with these commutators the Hamiltonian takes the form

$$H = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \sum_{s=1,2} (a^{(s)\dagger}(\mathbf{k}) a^{(s)}(\mathbf{k}) - b^{(s)\dagger}(\mathbf{k}) b^{(s)}(\mathbf{k})),$$

up to vacuum terms.

- Explain clearly the two problems of this quantization scheme:

- (a) the statistics of the quanta,
- (b) the fact that the Hamiltonian is not bounded below.

Why do these problems force us to abandon commutators for the Dirac field?

Problem 2. Anticommutation Relations, Fock Space, Momentum, and Charge (35 points)

We now quantize the Dirac field using *anticommutators*. The equal-time canonical anticommutation relation is

$$\{\psi_\alpha(t, \mathbf{x}), \psi_\beta^\dagger(t, \mathbf{x}')\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{x}').$$

The field expansion is

$$\psi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \sum_{s=1,2} [a^{(s)}(\mathbf{k}) u^s(k) e^{ik \cdot x} + b^{(s)\dagger}(\mathbf{k}) v^s(k) e^{-ik \cdot x}].$$

- Derive the operator anticommutation relations

$$\{a^{(s)}(\mathbf{k}), a^{(r)\dagger}(\mathbf{k}')\} = (2\pi)^3 \delta^{sr} \delta^{(3)}(\mathbf{k} - \mathbf{k}'),$$

$$\{b^{(s)}(\mathbf{k}), b^{(r)\dagger}(\mathbf{k}')\} = (2\pi)^3 \delta^{sr} \delta^{(3)}(\mathbf{k} - \mathbf{k}'),$$

with all other anticommutators vanishing.

- Define the vacuum by

$$a^{(s)}(\mathbf{k})|0\rangle = 0, \quad b^{(s)}(\mathbf{k})|0\rangle = 0.$$

Show that the Hamiltonian becomes

$$H = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \sum_{s=1,2} [a^{(s)\dagger}(\mathbf{k}) a^{(s)}(\mathbf{k}) + b^{(s)\dagger}(\mathbf{k}) b^{(s)}(\mathbf{k})] + E_0,$$

and explain why this is now bounded from below.

- Using the algebra

$$\{a, a^\dagger\} = 1, \quad \{a, a\} = 0, \quad \{a^\dagger, a^\dagger\} = 0,$$

show that

$$(a^\dagger)^2 = 0.$$

Explain how this implies the Pauli exclusion principle for each fermionic mode.

- Define one-particle and one-antiparticle states by

$$|\mathbf{k}, s\rangle = \sqrt{2\omega_{\mathbf{k}}} a^{(s)\dagger}(\mathbf{k})|0\rangle, \quad |\mathbf{k}, \bar{s}\rangle = \sqrt{2\omega_{\mathbf{k}}} b^{(s)\dagger}(\mathbf{k})|0\rangle.$$

Compute their inner products and show that particle and antiparticle states are orthogonal.

- Starting from

$$\mathbf{P} = \int d^3x \psi^\dagger (-i\nabla) \psi,$$

show that

$$\mathbf{P} = \int \frac{d^3k}{(2\pi)^3} \mathbf{k} \sum_{s=1,2} [a^{(s)\dagger}(\mathbf{k})a^{(s)}(\mathbf{k}) + b^{(s)\dagger}(\mathbf{k})b^{(s)}(\mathbf{k})].$$

Verify that the one-particle and one-antiparticle states are eigenstates of momentum with eigenvalue $\mathbf{k}$.

- The Dirac action is invariant under the global $U(1)$ transformation

$$\psi \rightarrow e^{i\alpha} \psi, \quad \bar{\psi} \rightarrow e^{-i\alpha} \bar{\psi}.$$

The associated Noether current is

$$j^\mu = -\bar{\psi} \gamma^\mu \psi.$$

Show that the conserved charge

$$Q = \int d^3x \psi^\dagger \psi$$

takes the form

$$Q = \int \frac{d^3k}{(2\pi)^3} \sum_{s=1,2} [a^{(s)\dagger}(\mathbf{k})a^{(s)}(\mathbf{k}) - b^{(s)\dagger}(\mathbf{k})b^{(s)}(\mathbf{k})] + Q_0,$$

up to a vacuum constant Q_0 . Explain why particles and antiparticles carry opposite

charge.